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**Pulse Student: An
Integrated Platform for
Grievance Redressal
Academic Support and
Smart Campus
Collaboration**

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Introduction



In educational institutions, students face multiple daily issues such as poor hostel management, mess food complaints, Wi-Fi problems, and academic disturbances. These problems often remain unresolved due to the absence of a systematic complaint management process.

Pulse Student is an innovative platform designed to empower students to raise their concerns, support existing issues through feedback and prioritization, and assist the administration in identifying and addressing the most critical problems efficiently, ensuring a better campus experience for all.



Education to Eternity

Literature Review



S. No.	Title of the Article Author Year of publication	Focus of Study, Design, Objectives , Method used and Sample size	Findings of the study and their conclusions	Research Gap
1	Decision-Making and Satisfaction in Campus E-Voting: Moderating Effect of Trust in the System (2022)	Focus: This study examines how electronic voting systems impact student decision-making and satisfaction in university elections, along with the role of trust in the system. Design: Survey-based study conducted in a university setting. Objectives: To analyze students' satisfaction with e-voting, understand the impact of trust, and assess its effectiveness in increasing participation.	• Findings: Students prefer e-voting due to its speed, convenience, and accessibility. Trust plays a major role in participation, although concerns about security and fairness still exist.	• The study is limited only to student elections and does not explore broader applications like hostel management, student complaints, event planning, video conferencing for discussions, or sharing important resources among students.



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Literature Review



S. No.	Title of the Article Author Year of publication	Focus of Study, Design, Objectives , Method used and Sample size	Findings of the study and their conclusions	Research Gap
2	Smart Campus Online Voting System (2023)	• Focus: Development of a secure online voting system to streamline and make the campus election process transparent and efficient. Design: Prototype-based development. Objectives: To create a secure, user-friendly voting platform with multi-layer authentication and fraud prevention measures.	Findings: The system successfully provided secure and efficient voting, saved time, improved convenience, and prevented fraudulent activities.	The system is focused only on elections. It does not include additional features like problem voting, complaint management, video meetings for student discussions, or resource sharing for collaborative solutions.



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Literature Review



S. No.	Title of the Article Author Year of publication	Focus of Study, Design, Objectives , Method used and Sample size	Findings of the study and their conclusions	Research Gap
3	Online Complaint Management System in Educational Institutions (2023)	Focus of Study, Design, Objectives, Method Used, and Sample Size: Focus: To design a centralized online system to manage student complaints efficiently within universities. Design: Web-based portal prototype. Objectives: To simplify the process of complaint submission, tracking, and resolution. Method: System development with admin and student login.	Findings: The system successfully reduced manual handling of complaints, improved response time, and created transparency between students and the administration.	Limited to complaint submission and status tracking only; lacks prioritization through feedback/voting, video meetings, and resource sharing functionalities which are addressed in the Pulse Student system.



Problem Statement



In educational institutions, students frequently encounter unresolved issues related to hostel management, poor quality of mess food, Wi-Fi connectivity problems, academic disturbances, and other campus-related challenges. Traditional complaint systems are often inefficient, lack transparency, and fail to prioritize the most pressing student concerns. Additionally, there is no centralized platform where students can collectively raise issues, vote on common problems, or actively participate in decision-making processes.

Existing solutions, such as online voting systems, focus primarily on student elections but do not address everyday campus problems or encourage collaborative problem-solving. Therefore, there is a need for a unified platform that not only allows students to post and vote on issues but also provides features like video meetings for open discussions and resource sharing for better coordination. This will enable the administration to identify, prioritize, and resolve student problems efficiently based on collective feedback and participation.



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Objective



1.Primary Objective:

- ❑ **Effective Student Issue Prioritization:** Design and implement a digital platform that allows students to raise common issues (such as hostel management, mess food complaints, Wi-Fi problems, and academic disturbances) and enables voting to prioritize these issues based on majority needs, ensuring efficient resolution by the administration.

2.Secondary Objectives:

- ❑ **System Evaluation:** Compare the platform with traditional complaint systems to measure improvements in problem resolution and student satisfaction.
- ❑ **Automation:** Automate issue collection and tracking to reduce manual work and speed up resolutions.
- ❑ **Scalability:** Ensure the system handles various issue types across departments and student categories.
- ❑ **Collaboration:** Add features like video meetings and resource sharing to improve communication between students and administration.



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Scope of Project



Centralized Student Platform: A single hub for managing academic and social activities.

Resource Sharing: System for sharing and downloading PDFs, documents, and notes.

Events Management: Create, discover, and participate in campus events.

Communication Tools: Real-time notifications, email alerts, and SMS verification.

Role-based Access: Different access levels for students and admins.

Analytics & Dashboard: Admin dashboard for tracking platform usage and insights.

Responsive Design: Optimized interface for both mobile and desktop devices.

Future Expansion: Potential for mobile app, AI-powered recommendations, real-time chat, and video integration.

Multi-branch Support: Coverage for 20+ branches and year-wise content organization.

Enhanced Learning: Facilitates collaboration among students and improves academic performance.



Methodology



Step-by-Step Approach:

Step 1: Smart Problem Submission & Verification:

- Students submit issues with categories, descriptions, and proof (images/documents).
- AI-based duplicate issue detection to avoid spam and repeated complaints.
- Admin approval with auto-flagging of irrelevant or fake problems.

• Step 2: Dynamic Prioritization & Voting System:

- Real-time voting with weightage based on issue category and urgency.
- Trending problems highlighted on the dashboard with automated escalation.
- Vote-locking mechanism to prevent multiple votes from the same user.

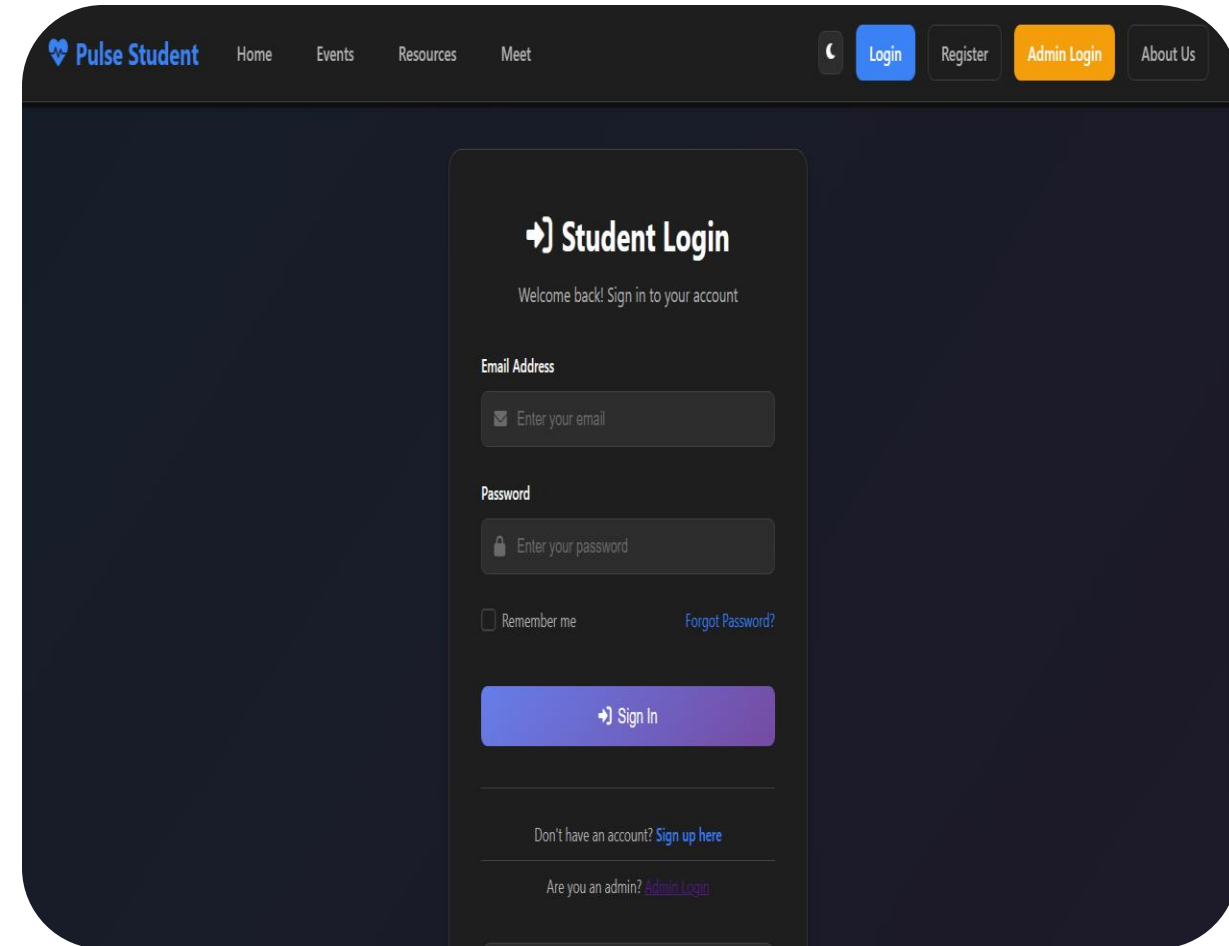
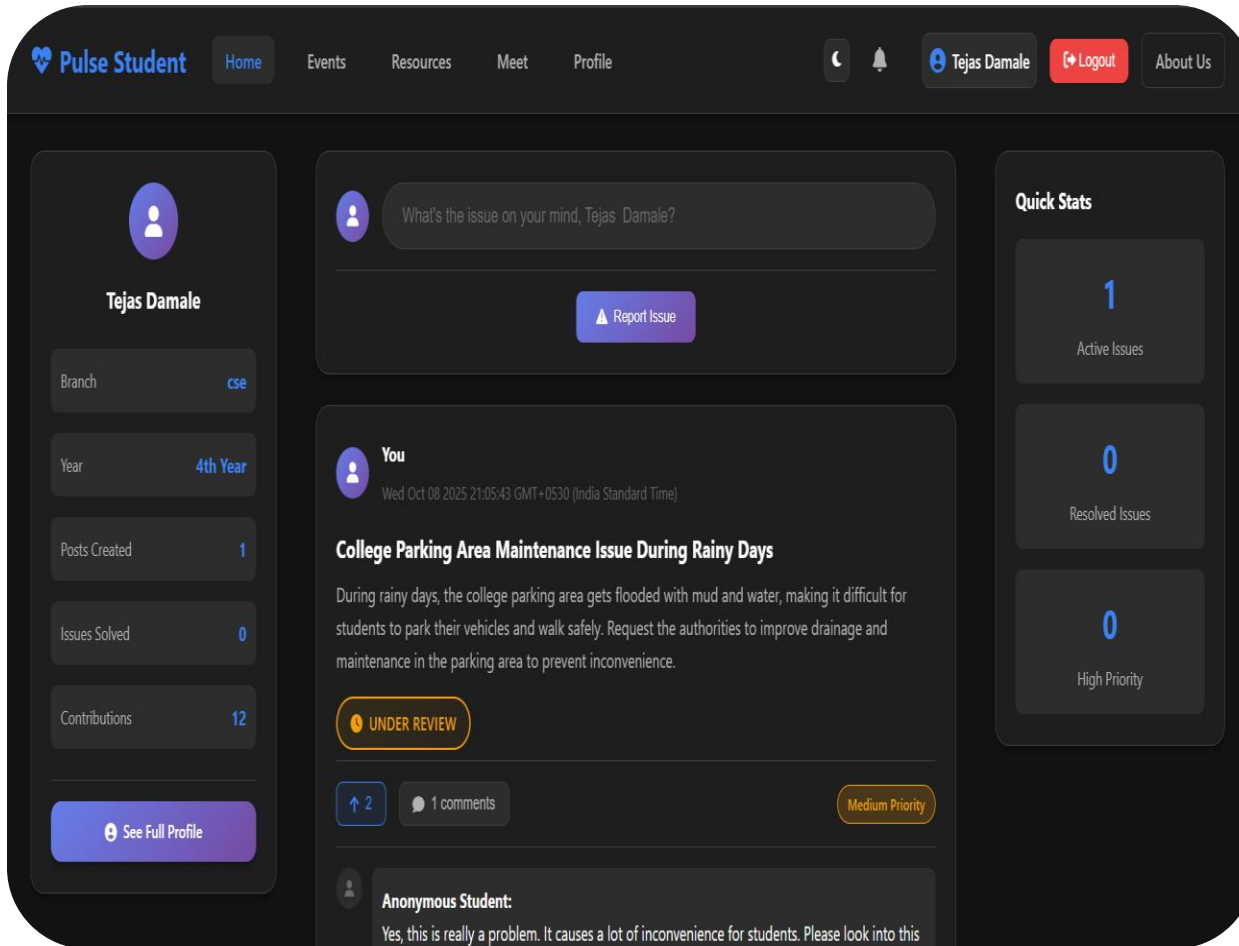
• Step 3: Resolution Management & Collaboration:

- Integrated video meetings for direct discussion between students and authorities.
- Resource sharing (PDFs, notices, guidelines) linked to each problem for transparency.
- Auto-status updates with feedback collection post-resolution.



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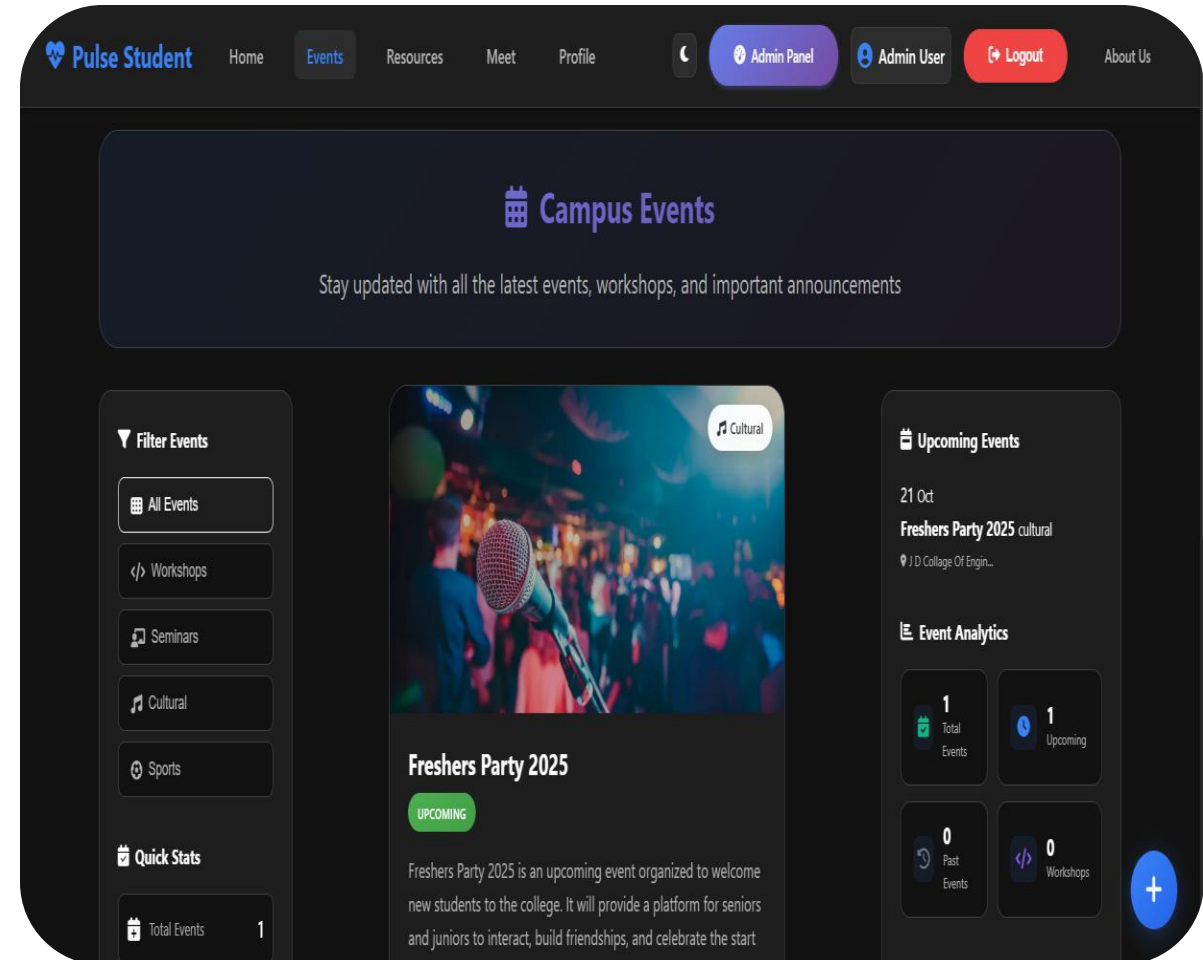
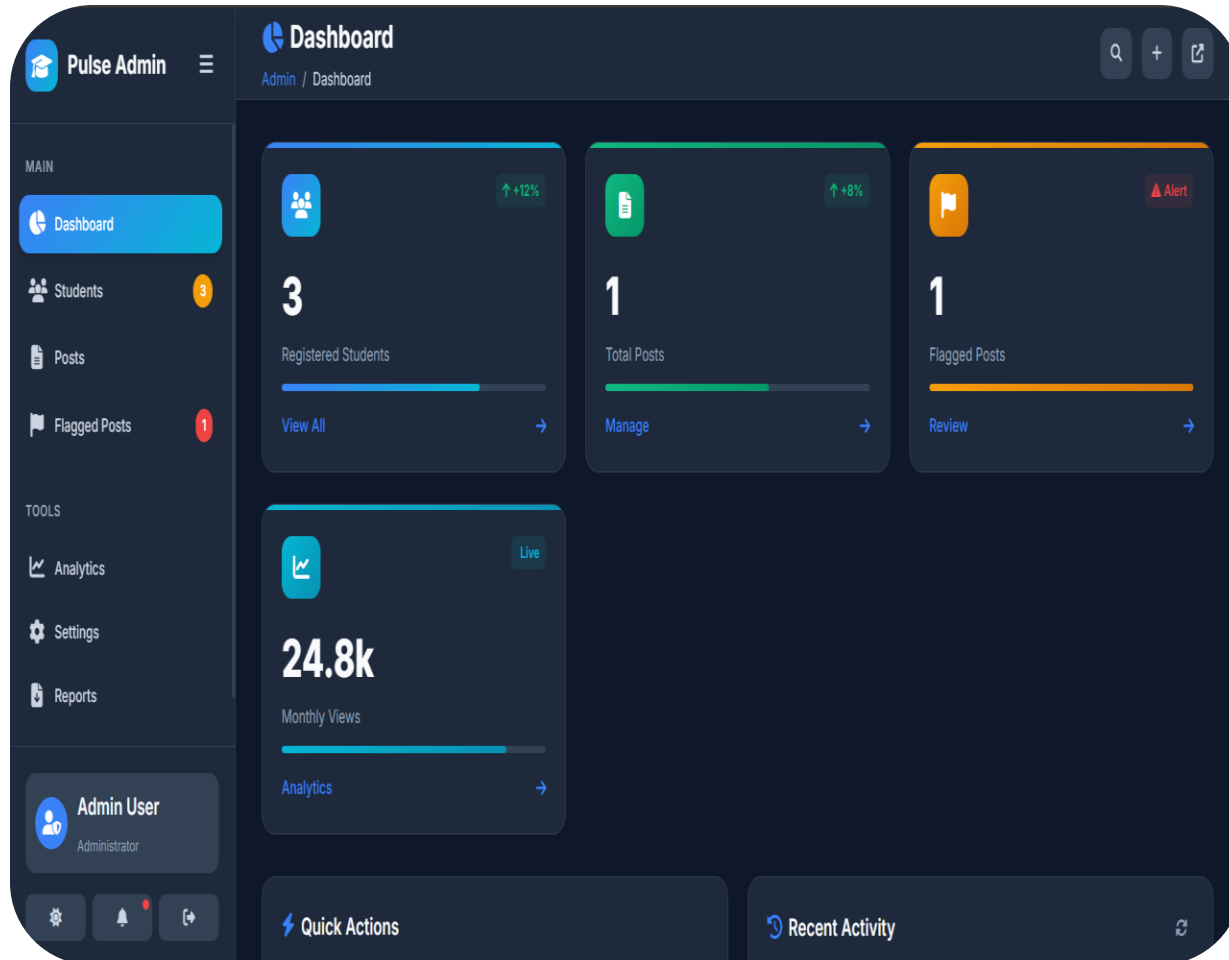
Project Screenshots





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Project Screenshots





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